
Beyond perplexity

Entropy-based evaluation of base language models

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Return to the language-modeling objective

Fixed benchmarks are useful slices, not a complete measure of a base model

Perplexity stopped comparing fairly

PPL is measured *per token*, but each model divides text differently: Llama-3.2 has **128,000** vocabulary entries and Qwen-2.5 has **151,643**.

BPB divides loss by UTF-8 bytes. A byte is the same amount of text for every model, so the scores share one unit.

Fixed benchmarks have blind spots

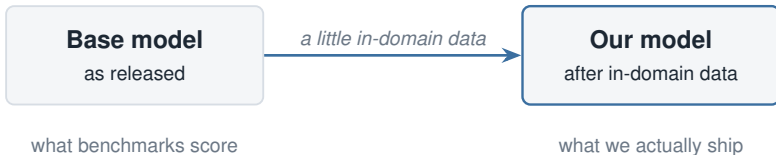
Task suites sample specific abilities, and repeated optimization on fixed tests makes contamination and benchmark-specific tuning harder to separate from general LM quality.

Entropy sees the full predictive distribution, but it needs a fair denominator, fresh target-domain text, and a contamination audit.

Go full circle: measure entropy again—with byte normalization, dynamic target-domain evaluation, controlled adaptation, and contamination filtering.

Measure the model we will actually use

A controlled adaptation turns evaluation into a deployment measurement



A benchmark describes the model as released. A product uses the model after domain data. We measure both states on the same held-out corpus.

The **adapted BPB level** selects the model. The reduction from zero-shot BPB explains how much calibration it needed.

Two measurements, one decision

1 — Use a fair ruler

Normalize cross-entropy onto *bytes* instead of tokens:

$$\text{BPB} = \frac{\mathcal{L}_{\text{nats}}}{\ln 2} \times \frac{|\text{tokens}|}{|\text{bytes}|}$$

UTF-8 bytes are the same unit for every model, so BPB is **tokenizer-agnostic**. Same physical text, same denominator, comparable numbers.

Not ours — The Pile, Paloma. Under-used, and we make it the primary metric.

2 — Measure domain fit

For each model, on one identical corpus:

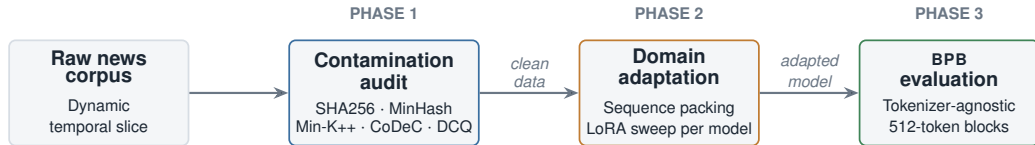
1. Zero-shot BPB
2. A *tuned* LoRA adaptation step
3. Adapted BPB

Primary output: adapted BPB for model selection.

Diagnostic: relative reduction for corpus calibration.

The pipeline

Chronologically strict: nothing is scored before it clears the gate



Corpus: one deduplicated slice of Google News, 8 June 2026 (119k articles), split 80/10/10 with zero intersection between splits. **Cohort:** 11 base models, 0.5B–35B, dense Transformers + a sparse MoE + a Liquid architecture.

Phase 1: why we trust the numbers

“Scout, then verify” — cheap detectors propose, an expensive test decides

- **Min-K++** (scout) — inspect the least-likely 20% of tokens; high confidence on rare tokens looks like recall, not prediction.
- **CoDeC** (scout, ours) — supply the preceding context; memorized text shows a non-linear *spike* in confidence.
- **Dynamic Choice Quiz** (gatekeeper) — hide the real passage among 16 synthetic near-duplicates. If the model reliably picks the original, quarantine it.

Run **per model**; we quarantine the **union** of every model’s verified leaks.

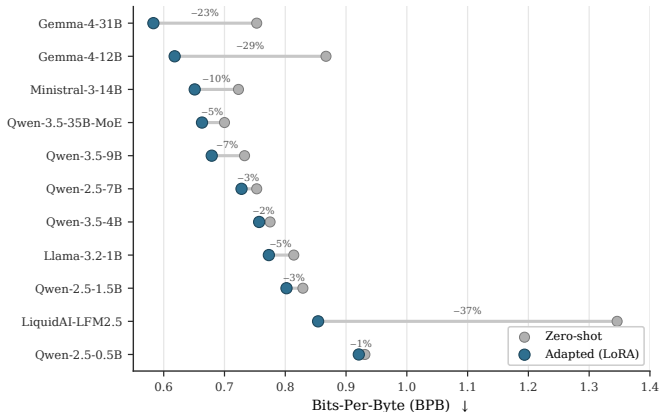
⇒ all 11 models are adapted and scored on **one identical, cleansed corpus**.

80 sequences removed globally.

This rules out benchmark memory as the explanation for an adaptation gain.

Results: 11 models, one ruler

Grey = as released, blue = after adaptation. Left is better.



Gemma-4-31B reaches the best adapted BPB: **0.583**.

The **35B MoE** lands with dense 12–14B models, giving a direct quality measurement between its active and total parameter counts.

LFM-2.5 shows the largest calibration shift: **36.5%**.

Selection uses the blue endpoint. Line length explains how much each base distribution moved toward the corpus.

Two numbers, two jobs

The level chooses a model; the reduction explains calibration

Adapted BPB

The primary selection score: how efficiently the adapted model predicts held-out target-domain text.

Lower is better. It is directly comparable across tokenizers and architectures.

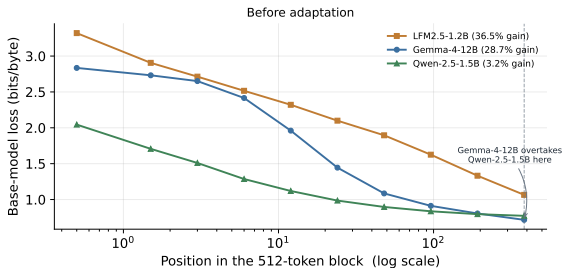
Relative reduction

A diagnostic: how much the released model recalibrates to the corpus's style, formatting, and context pattern.

It explains the trajectory; it does not replace the adapted endpoint.

Token-level evidence makes that interpretation visible.

Context explains the calibration signal



Corpus loss is aggregated over independent 512-token blocks, so nearly all cuts start mid-document with *no preceding context*.

The byte-normalized curves cross.

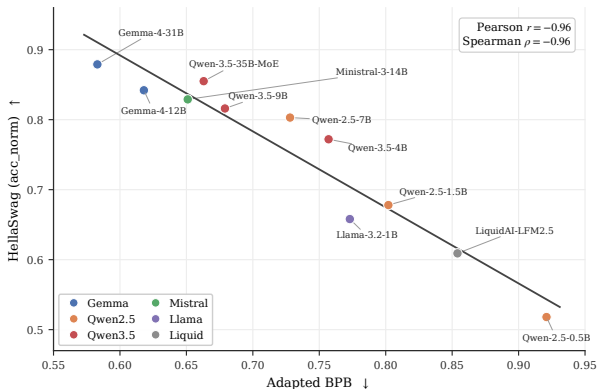
Over the first 8 positions Gemma averages 2.566 bits/byte vs Qwen's 1.490 — but at position 256+ Gemma is **0.717** and Qwen **0.773**.

The crossing survives tokenizer normalization: vocabulary size alone does not explain it. With context, the 12B model shows the stronger steady-state result.

Among the top 200 gain-carrying tokens: ':', ',', '\n\n', 'the', 'and'. Formatting, punctuation, and function words carry 83% of that mass.

BPB agrees when tasks measure language

HellaSwag provides an independent sentence-continuation check



Adapted BPB vs. HellaSwag: $r = -0.96$.

HellaSwag is sentence *continuation* — the benchmark closest to plain language modelling. There, BPB reproduces it almost exactly.

The result confirms that BPB tracks language-modelling quality using one corpus we control and one unit shared by every model.

MMLU-Pro and GSM8K then add complementary views of knowledge and arithmetic.

Different tasks reveal different strengths

▲*n*/ ▼*n* = ranks *n* places higher/lower than BPB puts it

Model	Adapted BPB <i>continuation</i>	HellaSwag <i>continuation</i>	MMLU-Pro <i>knowledge</i>	GSM8K <i>arithmetic</i>
Gemma-4-31B	1	1 ●	2 ▼1	3 ▼2
Gemma-4-12B	2	3 ▼1	3 ▼1	6 ▼4
Ministral-3-14B	3	4 ▼1	5 ▼2	5 ▼2
Qwen-3.5-35B-MoE	4	2 ▲2	1 ▲3	1 ▲3
Qwen-3.5-9B	5	5 ●	4 ▲1	2 ▲3
Qwen-2.5-7B	6	6 ●	6 ●	4 ▲2
Llama-3.2-1B	7	7 ●	9 ▼2	9 ▼2
LiquidAI-LFM2.5	8	8 ●	8 ●	7 ▲1
Qwen-2.5-0.5B	9	9 ●	7 ▲2	8 ▲1
Ranks identical to BPB	—	6 / 9	2 / 9	0 / 9
Largest displacement	—	2	3	4

Agreement changes **monotonically** as the task moves from continuation to knowledge to arithmetic. Each column measures a different capability.

The family pattern explains the shift

GSM8K emphasizes arithmetic strengths learned during pre-training

Every **Qwen** model ranks **higher** on GSM8K than in the BPB ordering — +3, +3, +2, +1.

Every **Gemma / Mistral / Llama** model moves the other way — -2, -4, -2, -2.

With **zero exceptions**, the pattern reflects a clear difference in pre-training emphasis: GSM8K rewards arithmetic experience.

BPB keeps **language modelling** separate from **arithmetic skill**, so the two measurements can be used together.

The ninth model, LFM-2.5, also rises one place.

One pair, two capabilities

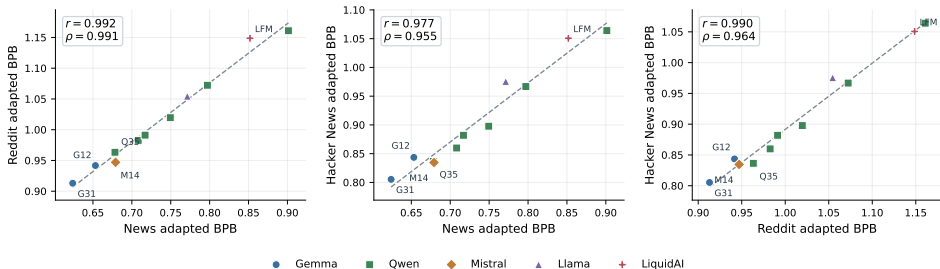
	Qwen-2.5-0.5B	Llama-3.2-1B
Adapted BPB ↓	0.921	0.773
HellaSwag ↑	0.518	0.658
GSM8K ↑	0.347	0.065

Llama is the better language model on both BPB and HellaSwag. On GSM8K the 0.5B Qwen scores **5×** **higher**.

The preferred model follows the capability the product needs.

Adaptation changes the ranking — then it transfers

Within-domain reordering; cross-domain stability after adaptation



Adaptation changes selection:

zero-shot $\rightarrow$ adapted $\rho = 0.655/0.609/0.682$;
maximum moves are **7/7/5 places**.

Adapted ranks transfer: cross-domain

$\rho = +0.991/ +0.955/ +0.964$; no model moves
more than two places.

Gemma-4-31B moves from zero-shot rank **4/5/6** to adapted rank **1/1/1** on news, Reddit, and Hacker News.

A simple decision rule

What we learned

- One coordinate system that is valid across tokenizers and architectures.
- Adaptation can change the choice; the adapted ranking remains stable across three corpora.
- A 512-token adapter retains its gain through 2,048-token inference.

How to use it

- Collect recent text from the target domain.
- Clean and audit it, then adapt every candidate under one budget.
- Rank by adapted BPB and test near-ties on the target domain.

Bottom line: adapted BPB gives one fair ruler, survives a three-domain test, and turns a fresh corpus into a defensible base-model decision.

Thank you

Questions?

Backup slides follow: full benchmark scores, contamination breakdown, adaptation protocol, longer-context transfer.

Backup: the adaptation results as a table

Lower BPB is better. The reduction is a calibration diagnostic, not plasticity — see slide 9.

Model	Architecture	Params	Zero-shot	Adapted	Reduction
Gemma-4-31B	Dense Transformer	31B	0.753	0.583	22.6%
Gemma-4-12B	Dense Transformer	12B	0.867	0.618	28.7%
Ministral-3-14B	Dense Transformer	14B	0.723	0.651	9.9%
Qwen-3.5-35B-MoE	Mixture of Experts	35B	0.700	0.663	5.4%
Qwen-3.5-9B	Dense Transformer	9B	0.733	0.679	7.4%
Qwen-2.5-7B	Dense Transformer	7B	0.753	0.728	3.3%
Qwen-3.5-4B	Dense Transformer	4B	0.775	0.757	2.3%
Llama-3.2-1B	Dense Transformer	1.2B	0.814	0.773	5.1%
Qwen-2.5-1.5B	Dense Transformer	1.5B	0.829	0.802	3.2%
LiquidAI-LFM2.5	Liquid Neural Net	1.2B	1.346	0.854	36.5%
Qwen-2.5-0.5B	Dense Transformer	0.5B	0.931	0.921	1.0%

Every value here reproduces from `results/summary.json` at full precision.

Backup: the full numbers behind slides 11–12

The ranks and displacements on slide 11 are derived from this table

Model	Adapted BPB	MMLU-Pro	HellaSwag	GSM8K	Macro
Gemma-4-31B	0.583	0.612	0.865	0.817	0.764
Gemma-4-12B	0.618	0.583	0.840	0.778	0.733
Ministral-3-14B	0.651	0.541	0.829	0.807	0.726
Qwen-3.5-35B-MoE	0.663	0.620	0.855	0.865	0.780
Qwen-3.5-9B	0.679	0.562	0.816	0.860	0.746
Qwen-2.5-7B	0.728	0.475	0.803	0.813	0.697
Llama-3.2-1B	0.773	0.117	0.658	0.065	0.280
LiquidAI-LFM2.5	0.854	0.155	0.609	0.550	0.438
Qwen-2.5-0.5B	0.921	0.167	0.518	0.347	0.344

Rank analysis uses the nine-model common subset reported across BPB and all three tasks. Harness: *lm-evaluation-harness*, vLLM backend, bf16, greedy decoding, each tokenizer's native BOS. MMLU-Pro 5-shot CoT on a fixed 1,000-question subset; HellaSwag 10-shot length-normalized; GSM8K 5-shot flexible-extract.

Backup: contamination breakdown

$N = 1000$ sampled sequences per model; 80 unique global leaks removed

Model	Min-K++ flags	CoDeC flags	DCQ candidates	DCQ acc.	Verified leaks
google-gemma-4-31B	12	6	20	100%	20
google-gemma-4-12B	11	6	11	100%	11
mistralai-Mistral-3-14B	12	8	13	100%	13
Qwen-Qwen3.5-35B-MoE	6	8	11	100%	11
Qwen-Qwen3.5-9B	12	6	17	100%	17
Qwen-Qwen2.5-7B	10	6	14	100%	14
Qwen-Qwen3.5-4B	9	6	12	100%	12
meta-llama-Llama-3.2-1B	17	0	28	100%	28
Qwen-Qwen2.5-1.5B	14	4	17	100%	17
LiquidAI-LFM2.5	12	8	23	96%	22
Qwen-Qwen2.5-0.5B	8	2	13	100%	13

Model size does *not* linearly predict memorization — Llama-3.2-1B leaks most in this cohort.

Backup: adaptation protocol

Data handling

- Google News slice, 8 June 2026: 119,054 articles, ~ 361 MB.
- 80/10/10 train/val/test, seed 42, enforced zero intersection between splits.
- Packed into contiguous 512-token blocks along EOS boundaries.

Scoring

- Corpus-wide cross-entropy over independent 512-token test blocks.
- Context resets at block cuts; injected start markers are context, not targets.
- Model-specific bytes-per-token factor estimated from the test text.

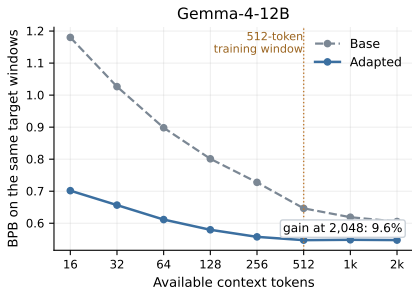
Per-model HPO

- 20-trial Optuna sweep, Successive Halving (top 33% promoted).
- LoRA rank $r \in \{8, 16, 32, 64\}$; α -multiplier $\in \{1, 2, 4\}$; dropout; LR 10^{-6} – 10^{-3} log-scale; effective batch to 128 via gradient accumulation.
- Adapters restricted to the language backbone.
- Winner trained to convergence: cosine schedule to $0.1 \times$ peak LR, gradient clip 1.0.

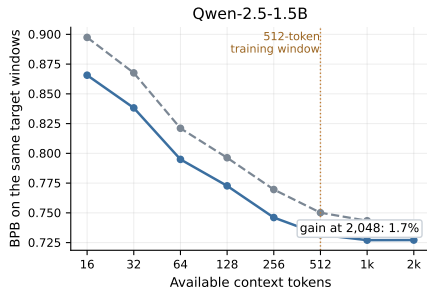
Hardware: A100/H100 multi-node under Slurm.

Backup: 512-token adaptation transfers to longer contexts

Same held-out targets; only the available preceding context changes



Gemma base BPB keeps recovering after 512:
0.6465 → **0.6054** at 2,048 tokens.



Adapted Gemma stays at **0.547**; its advantage remains **9.6%** at four times the training context.

The adapter does not require the inference sequence to stop at 512 tokens.